Math 4720: Statistical Methods

9^{th} Week Summary (10/29/26)

- **NON-PARAMETRIC TESTS** ....

- Test for the population median M:

  SIGN TEST:

  Test Statistics: B = # of data values $> m_0$

  Minitab: Stat → Nonparametrics → 1-Sample Sign

  R: library("BSDA"); SIGN.test(dataset)

- Non-parametric TWO Independent Samples Test:

  WILCOXON RANK-SUM TEST(MANN–WHITNEY U TEST)

  Combine both the samples. Rank all values of the combined sample from lowest to the highest.

  T= sum of the ranks insample 1

  Minitab: Stat → Nonparametrics → Mann-Whitney

  R: wilcox.test(group1, group2)

- Non-parametric TWO Dependent Samples Test:

  WILCOXON SIGNED-RANK TEST

  Test Statistics: Rank the absolute values of the differences.

  T= sum of the ranks of negative(positive) differences

  Minitab: For differences, Minitab: Stat → Nonparametrics → 1-Sample Wilcoxon

  R: wilcox.test(before, after, paired = T)

- **INFERENCE ABOUT POPULATION STANDARD DEVIATION** ....

- Test for the population standard deviation (σ):

  Given the data is coming from normal distribution we can use the chi-squared test to test $H_0 : \sigma = \sigma_0$:

  Test Statistics: $\chi^2 = \dfrac{(n-1)s^2}{\sigma_0^2}$ follows $\chi^2(\nu = n-1)$

  $100(1-\alpha)\%$ CI for σ: $\sqrt{\dfrac{(n-1)s^2}{\chi^2_{\alpha/2}}} < \sigma < \sqrt{\dfrac{(n-1)s^2}{\chi^2_{1-\alpha/2}}}$

  R: EnvStats::varTest(data, sigma.squared = 1)

- Test for the equality of the standard deviations ($H_0 : \sigma_1 = \sigma_2$):

  Given the data-sets are coming from normal distribution we can use the F test to test $H_0 : \sigma_1 = \sigma_2$

  Test Statistics: $F = \dfrac{\max(s_1^2, s_2^2)}{\min(s_1^2, s_2^2)}$ follows $F(\nu_1 = n_{num} - 1, \nu_2 = n_{den} - 1)$

  R: var.test(x, y, ratio = 1)

- Test for equality of the standard deviations for more than two populations ($H_0 : \sigma_1 = \sigma_2 = \cdots = \sigma_t$):

  F test can be extended to more than two populations (Hartley's $F_{\max}$ test), but it is very sensitive to departures from normality.

  We prefer to use Brown-Forsythe-Levene(BFL) test:

  R: lawstat::levene.test(data, group)